

LLMOps for Drug Discovery Systems

Governing Models, Agents, and Evaluation

Foundation models and AI agents are transforming drug discovery, but real-world deployment demands strong MLOps and LLMOps. Using Weights & Biases Models and Weave, we demonstrate a reproducible workflow for fine-tuning, agent tracing, evaluation, and RL that improves reliability, speeds iteration, and supports compliance.

BioIT World • 22-minute talk

Talk Structure (22 min)

1. Case for a systematic improvement loop (6 min)
2. W&B tools and what they unlock (5 min)
3. Our BioReason-Pro plan: ablations + tuning strategies (6 min)
4. Results + live W&B demo (5 min)

Slides for narrative + live UI for evidence.

1) Why We Need a Systematic Loop

- AI systems now generate hypotheses, candidates, and rankings.
- But many pipelines are still stitched together with scripts and notebooks.
- That breaks reproducibility, governance, and learning speed.

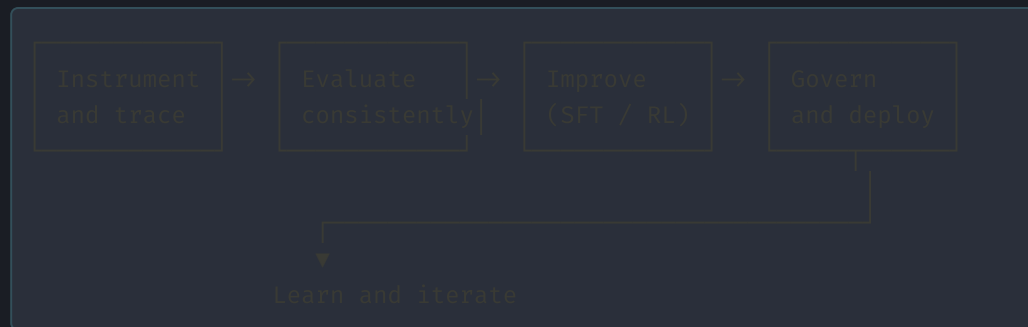
Building models is no longer enough; operating them is the real bottleneck.

What "Ad Hoc" Looks Like

```
model_v1 → notebook_eval → spreadsheet_metrics → manual notes → model_v2
|           |           |           |
unclear    non-repeatable    no lineage    no audit trail
```

- Which exact run produced this claim?
- Why did performance shift this week?
- Is this gain real, or split leakage / randomness?

Systematic Improvement Loop



The point is not one good run; the point is a reliable learning system.

2

W&B Tools and How They Help

Models + Weave + Evaluations as the operating layer

W&B Models: Versioning and Lineage

- Register base models, tuned checkpoints, and aliases.
- Track datasets, configs, and artifacts per run.
- Promote with explicit criteria (not intuition).

```
model: bioreason-pro  
alias: pilot-disease-go-v3  
parent: bioreason-pro-base  
training_data: disease-human-go-v2  
eval_gate: temporal_holdout_go_v1 ≥ target
```

Weave: Traces for Agents and Model Calls

- Capture each call, intermediate step, latency, and output.
- Debug failed predictions with full context.
- Compare behavior across versions and prompt templates.

```
@weave.op
def go_function_agent(sample):
    # trace input → retrieval → reasoning → output
    return run_agent(sample)
```


Evaluation: Measuring Real Improvement

same split + same metrics + same protocol $\rightarrow$ fair comparison

- Evaluate by biologically meaningful metrics and slices.
- Track confidence intervals and temporal holdout behavior.
- Decide promotions with explicit eval gates.

3

What We're Doing: Pilot Plan

BioReason-Pro pilot on disease-related human proteins for GO function prediction

- Objective: improve GO prediction quality.
- Constraint: **no architecture changes**.
- Focus: data strategy + tuning strategy + evaluation rigor.

Ablation Grid (Planned)

Axes:

1) Data scope

- generic proteins
- + disease-related human proteins
- + curated hard cases

2) Training strategy

- baseline (no tune)
- SFT
- RL (GRPO-style variants)

3) Prompt / context strategy

- base prompt
- ontology-aware prompt
- retrieval-augmented context

Run each cell with the same eval suite for apples-to-apples comparisons.

Tuning Strategies We Are Testing

Baseline

Current BioReason-Pro, no tuning.

SFT

Supervised fine-tuning on curated disease-protein examples.

RL Variant A

Reward prioritizes GO correctness + calibration.

RL Variant B

Reward adds robustness penalties for known failure modes.

Execution Plan

```
Week 1: freeze baselines + finalize eval split  
Week 2: run SFT ablations  
Week 3: run RL variants + reward sensitivity  
Week 4: compare, select, and package candidate model
```

- Every run logged in W&B with linked artifacts.
- Promotion candidate requires passing predefined eval gates.
- Keep all decisions traceable for technical and compliance review.

4

Results + Live Demo

Slides for summary, W&B UI for evidence

Results Snapshot (Replace with latest)

TBD

delta Fmax

TBD

delta AUPRC

TBD

best strategy

0

arch changes

- Show confidence intervals on final version of this slide.
- Keep metric definitions visible in backup slides.

Live W&B Demo: What I Will Show

1. Run comparison dashboard (baseline vs SFT vs RL variants)
2. Trace deep dive for one success and one failure case
3. Model registry lineage from base model to candidate
4. Eval report + promotion gate decision

Goal: show that every claim in slides is backed by inspectable evidence.

Closing Message

- Drug discovery AI needs model innovation **and** operating discipline.
- A systematic loop turns one-off experiments into cumulative progress.
- W&B tooling helps teams trace, evaluate, tune, and govern at production pace.
- Our pilot asks a focused question and tests it with reproducible evidence.

If it cannot be reproduced and explained, it cannot be trusted.

Questions

LLMOps for Drug Discovery Systems: Governing Models, Agents, and Evaluation